

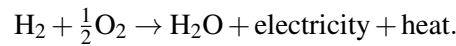
# Short Report on PEM Fuel Cell Modeling

a brief overview of the 2 weeks work

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## 1. Introduction

A proton exchange membrane fuel cell (PEMFC) converts the chemical energy of hydrogen and oxygen into electricity, while also producing heat and water [1, 5, 2]. At the anode, hydrogen splits into protons and electrons. The protons pass through the polymer membrane, while the electrons move through the external circuit and deliver electrical power. At the cathode, oxygen combines with the protons and electrons to form water [1, 5]. The overall reaction is



The membrane is the core element of the PEMFC because it conducts protons, blocks electrons, and separates the fuel side from the oxidant side [1, 6]. Together with the anode and the cathode, it forms the membrane electrode assembly (MEA) [1, 7]. Water management is also essential for PEMFC operation. If the membrane is too dry, proton conductivity drops and both performance and durability decrease. If too much water accumulates, especially at the cathode, flooding can block gas transport to the reaction sites and reduce performance. In practice, PEMFC operation requires a balance: enough water to keep the membrane hydrated, but not so much that reactant transport is hindered [7, 6, 2, 8].

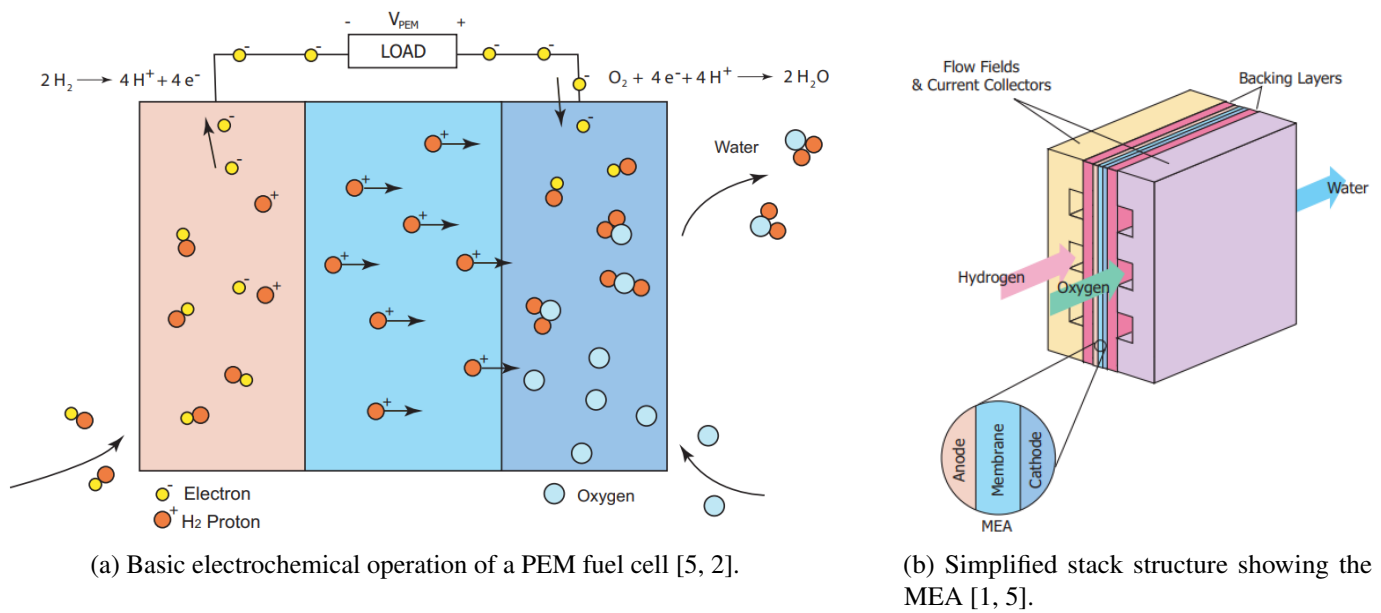


Figure 1: Operating principle and physical structure of a PEM fuel cell.

A practical fuel cell does not deliver the ideal reversible voltage because some of the available electrochemical energy is lost during operation. The three main voltage losses are activation loss, ohmic loss, and concentration loss [1, 4, 3]. They come from different physical causes and tend to dominate in different parts of the polarization curve.

- **Activation loss** is the extra voltage needed to make the electrode reactions proceed on the catalyst surface. It is mainly a reaction-kinetics limitation [4, 6].
- **Ohmic loss** is the voltage drop caused by resistance to proton flow in the membrane and electron flow in the solid parts of the cell. It usually increases roughly linearly with current [4, 3, 6].
- **Concentration loss** occurs when reactants cannot reach the reaction sites fast enough, especially at high current density. It is mainly a mass-transport limitation and causes a sharp voltage drop at high load [1, 4, 6].

In short, activation loss is a **kinetic** limitation, ohmic loss is a **resistive** limitation, and concentration loss is a **transport** limitation.

This is why the cell voltage decreases when the current density increases, as shown by the typical polarization curve in Figure 2b. At low current, activation effects dominate; in the middle region, ohmic loss gives the main voltage decrease; and at high current, concentration loss becomes severe and causes the strong final drop of voltage [4, 6]. Figure 2a illustrates separately the effect of activation, ohmic, and concentration losses, as well as their combined effect on the cell voltage. Higher operating pressure can improve reaction rate and power density, while the fast transient behavior of the cell is also influenced by the charge double-layer effect at the electrode/electrolyte interface, which behaves like a small capacitor in equivalent electrical models [4].

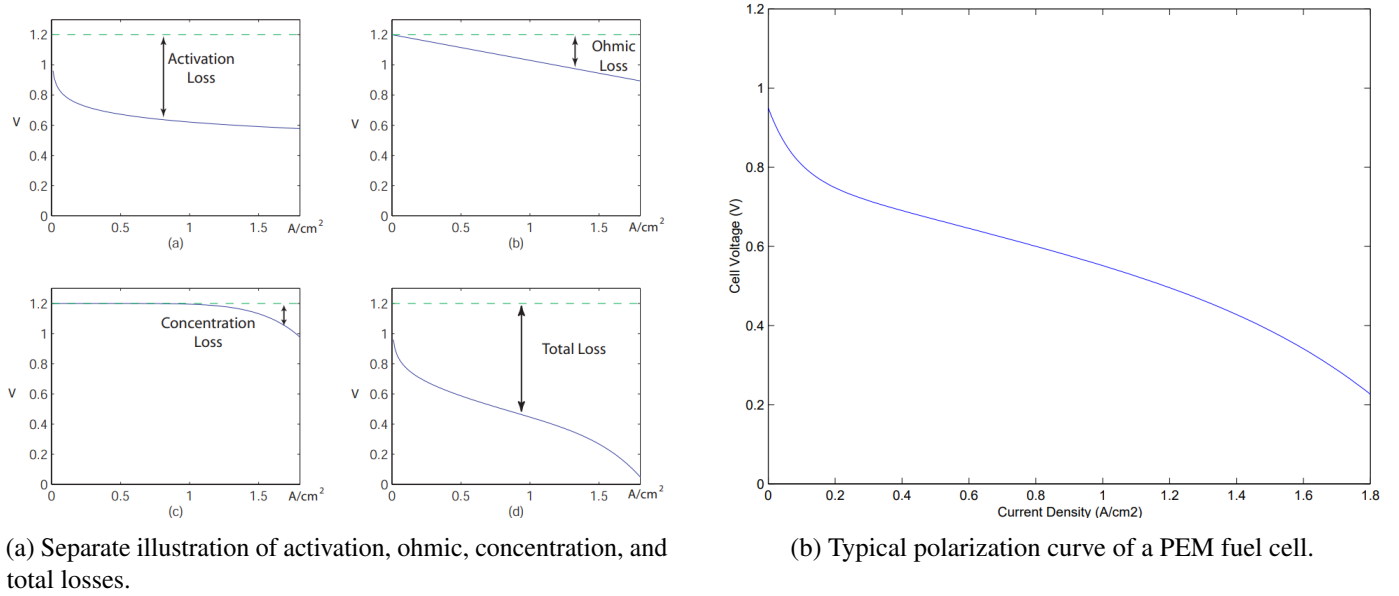


Figure 2: Influence of the main voltage losses on PEM fuel cell performance.

## 2. Reduced PEMFC Modeling

### 2.1. Modeling idea and assumptions

To study control and simulation problems, it is common to use a reduced-order model instead of a full three-dimensional multiphysics model [3, 6, 8]. In the reduced model used here, the fuel cell is decomposed into three interacting parts [3, 4]:

- 1) the hydrogen/anode side,
- 2) the membrane,
- 3) the oxygen/cathode side.

The main assumptions are the following [3, 5, 4]:

- hydrogen and air are the only reactants considered,
- pressure drops inside the channels are neglected,
- gases are treated with ideal-gas relations,
- temperature is taken as constant or slowly varying,
- the oxygen partial pressure is assumed constant and equal to  $0.21 P_{\text{atm}}$ ,
- each side is lumped into a single control volume.

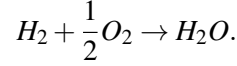
This kind of model is mainly simplified for control design, while still remaining useful for simulation and identification [3, 4]. However, it is important to state clearly that it is **not a fully physics-based distributed model**. It is better described as a **semi-empirical or reduced model**: the mass-balance structure is physically motivated, while some

voltage-loss relations are written in a simplified empirical form and can later be calibrated from experimental data if needed [4, 6, 9].

## 2.2. Fuel Cell Voltage

### 2.2.1 Open-circuit Voltage

The goal of this subsection is to show how the reversible PEMFC voltage is obtained from thermodynamics and how the numerical expression used later in the model is constructed. The derivation starts from the global hydrogen–oxygen reaction [4, 1]



At non-standard conditions, the Gibbs free energy change of this reaction is [4]

$$\Delta g_f = \Delta g_f^0 - \bar{R}T_{fc} \ln \left( \frac{p_{H_2} p_{O_2}^{1/2}}{p_{H_2O}} \right),$$

where  $\Delta g_f^0$  is the Gibbs free energy change at standard pressure,  $\bar{R}$  is the universal gas constant, and  $p_{H_2}$ ,  $p_{O_2}$ , and  $p_{H_2O}$  are the partial pressures of hydrogen, oxygen, and water vapor [4].

For a reversible fuel cell, the maximum electrical work is equal to the decrease in Gibbs free energy. Since one mole of hydrogen transfers two moles of electrons, the reversible electrical work satisfies [4, 1]

$$\Delta g_f = -2FE,$$

where  $F$  is Faraday's constant and  $E$  is the reversible open-circuit voltage. Therefore, the reversible voltage becomes [4]

$$E = -\frac{\Delta g_f}{2F} = -\frac{\Delta g_f^0}{2F} + \frac{\bar{R}T_{fc}}{2F} \ln \left( \frac{p_{H_2} p_{O_2}^{1/2}}{p_{H_2O}} \right).$$

This equation shows that the reversible voltage has two parts: a standard thermodynamic term,  $-\Delta g_f^0/(2F)$ , and a pressure-dependent correction given by the logarithmic Nernst term.

Table 1: Change in Gibbs free energy of hydrogen fuel cell at various temperatures [4].

| Form of water product | Temperature (°C) | $\Delta g_f^0$ (kJ/mol) |
|-----------------------|------------------|-------------------------|
| Liquid                | 25               | -237.2                  |
| Liquid                | 80               | -228.2                  |
| Gas                   | 80               | -226.1                  |
| Gas                   | 100              | -225.2                  |
| Gas                   | 200              | -220.4                  |
| Gas                   | 400              | -210.3                  |
| Gas                   | 600              | -199.6                  |
| Gas                   | 800              | -188.6                  |
| Gas                   | 1000             | -177.4                  |

To obtain the numerical constant used in the model, the standard liquid-water value at 25°C is taken as the reference [4]. At standard conditions, the logarithmic term vanishes, so the reference reversible voltage is

$$E^0 = -\frac{\Delta g_f^0}{2F} = -\frac{-237.2 \times 10^3}{2 \times 96485} \approx 1.229 \text{ V}.$$

Hence, the constant 1.229 is simply the standard reversible voltage of the hydrogen fuel cell at standard conditions [4].

The corresponding reference temperature is the standard-state temperature [4]

$$T_0 = 25^\circ\text{C} + 273.15 = 298.15 \text{ K},$$

To describe how the standard term changes with temperature, the thermodynamic identity

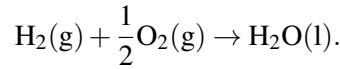
$$\Delta G^0 = \Delta H^0 - T\Delta S^0,$$

is used. Evaluated around the standard reference temperature  $T_0 = 298.15$  K, the temperature dependence of the standard reversible voltage can be written as [4]

$$-\frac{\Delta g_f^0}{2F} = 1.229 + (T_{fc} - T_0) \left( \frac{\Delta S^0}{2F} \right),$$

where 1.229 V is the standard reversible voltage at  $T_0 = 298.15$  K and standard pressure, and  $\Delta S^0$  is the standard entropy change of the overall fuel cell reaction.

So, the remaining step is to compute  $\Delta S^0$  for the reference reaction used in the model. If **liquid water** is taken as the reaction product, the global reaction is



In that case, the standard entropy change is obtained from the standard molar entropies of the reactants and products:

$$\Delta S^0 = S_{\text{H}_2\text{O}(\text{l})}^0 - S_{\text{H}_2(\text{g})}^0 - \frac{1}{2}S_{\text{O}_2(\text{g})}^0.$$

Using the standard thermodynamic values

$$\begin{aligned} S_{\text{H}_2(\text{g})}^0 &\approx 130.7 \text{ J mol}^{-1} \text{ K}^{-1}, \\ S_{\text{O}_2(\text{g})}^0 &\approx 205.15 \text{ J mol}^{-1} \text{ K}^{-1}, \\ S_{\text{H}_2\text{O}(\text{l})}^0 &\approx 69.95 \text{ J mol}^{-1} \text{ K}^{-1}, \end{aligned}$$

which gives

$$\Delta S^0 = 69.95 - 130.7 - \frac{1}{2}(205.15) \approx -163.3 \text{ J mol}^{-1} \text{ K}^{-1}.$$

Therefore,

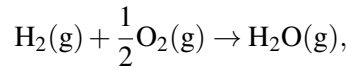
$$\frac{\Delta S^0}{2F} = \frac{-163.3}{2 \times 96485} \approx -8.46 \times 10^{-4} \text{ V K}^{-1} \approx -0.85 \times 10^{-3} \text{ V K}^{-1}.$$

After substituting this temperature correction and the pressure correction into the reversible-voltage expression, the final numerical form becomes [4]

$$E = 1.229 - 0.85 \times 10^{-3}(T_{fc} - 298.15) + 4.3085 \times 10^{-5} T_{fc} \left[ \ln(p_{\text{H}_2}) + \frac{1}{2} \ln(p_{\text{O}_2}) \right] \text{ V}, \quad (1)$$

where  $T_{fc}$  is the fuel cell temperature in Kelvin, and  $p_{\text{H}_2}$  and  $p_{\text{O}_2}$  are the hydrogen and oxygen partial pressures in atm [4]. The first term, 1.229, is the standard reversible voltage at 298.15 K; the second term is the temperature correction originating from the entropy change; and the last logarithmic term is the pressure correction coming from the Nernst relation.

It is important to note that the temperature coefficient above is valid only when **liquid water** is chosen as the reference product. If **water vapor** were used instead, then the reaction would be written as



and the standard entropy change would become

$$\Delta S_{\text{vap}}^0 = S_{\text{H}_2\text{O}(\text{g})}^0 - S_{\text{H}_2(\text{g})}^0 - \frac{1}{2}S_{\text{O}_2(\text{g})}^0.$$

Using

$$S_{\text{H}_2\text{O}(\text{g})}^0 \approx 188.84 \text{ J mol}^{-1} \text{ K}^{-1},$$

one finds

$$\Delta S_{\text{vap}}^0 \approx 188.84 - 130.7 - \frac{1}{2}(205.15) \approx -44.4 \text{ J mol}^{-1} \text{ K}^{-1},$$

which leads to a different value of  $\Delta S^0/(2F)$ . This is why the temperature coefficient in the reversible-voltage equation depends on whether the reaction product is taken as liquid water or water vapor.

### 2.2.2 Power Losses

In practice, the fuel cell voltage is lower than the reversible voltage given by (1) because the reaction is not ideal and several irreversible losses appear [1, 4, 3]. Thus, the stack voltage is written as the reversible open-circuit voltage minus the three major losses [3, 4]:

$$v_{fc} = E - v_{act} - v_{ohm} - v_{con}. \quad (2)$$

The activation loss is modeled by an empirical expression [4, 9]:

$$v_{act} = \xi_1 + \xi_2 T + \xi_3 T \ln(C_{O_2}) + \xi_4 T \ln(i_{FC}), \quad (3)$$

where  $i_{FC}$  is the operating current and  $\xi_1, \xi_2, \xi_3, \xi_4$  are fitted parameters.

The ohmic loss is written as [4, 3]

$$v_{ohm} = i_{FC} \left( \rho_m \frac{t_m}{A_{fc}} + c \right), \quad (4)$$

where  $\rho_m$  is the membrane resistivity,  $t_m$  is the membrane thickness,  $A_{fc}$  is the active area, and  $c$  is an additional fitted resistance term. The membrane resistivity  $\rho_m$  is the inverse of the membrane conductivity,

$$\rho_m = \frac{1}{\sigma_m}.$$

In the control-oriented PEMFC model, the membrane conductivity depends on both membrane hydration and temperature, and is written as [4]

$$\sigma_m = b_1 \exp \left[ b_2 \left( \frac{1}{303} - \frac{1}{T_{fc}} \right) \right],$$

with

$$b_1 = b_{11} \lambda_m - b_{12},$$

where  $\lambda_m$  is the membrane water content and  $b_{11}, b_{12}$ , and  $b_2$  are empirical constants. Therefore, the membrane resistivity can be written as

$$\rho_m = \frac{1}{(b_{11} \lambda_m - b_{12}) \exp \left[ b_2 \left( \frac{1}{303} - \frac{1}{T_{fc}} \right) \right]}. \quad (5)$$

The concentration loss is also represented empirically [4, 1]:

$$v_{con} = -B \ln(1 - \theta), \quad (6)$$

with

$$\theta = \frac{i_{FC}}{i_{\max}}$$

The parameters  $B$  and  $c$  are obtained from optimization, which again shows that the model is not purely first-principles [9].

### 2.3. Hydrogen model on the anode side

The anode model describes how the hydrogen stored in the anode changes with time and how this determines the hydrogen partial pressure [3, 4]. In the reduced control-oriented model, the anode is treated as a lumped control volume, so the gas properties are assumed uniform inside it.

The hydrogen partial pressure is obtained from the ideal-gas law:

$$P_{H_2} = \frac{m_{H_2} R_{H_2} T}{V_{an}},$$

where  $m_{H_2}$  is the hydrogen mass inside the anode,  $R_{H_2}$  is the specific gas constant of hydrogen,  $T$  is the operating temperature, and  $V_{an}$  is the anode volume [3, 4]. For fixed temperature and volume, the pressure rises when more hydrogen is stored and falls when hydrogen is removed or consumed.

The stored hydrogen mass follows the conservation law

$$\dot{m}_{H_2} = W_{H_2,in} - W_{H_2,out} - W_{H_2,reacted}.$$

This means that the hydrogen mass increases with the inlet flow, decreases with the outlet flow, and also decreases because part of the hydrogen is consumed by the electrochemical reaction [3, 4].

The consumption term is linked to the fuel cell current through Faraday's law:

$$W_{H_2,reacted} = \frac{M_{H_2}n}{2F}i_{FC},$$

where  $M_{H_2}$  is the molar mass of hydrogen,  $n$  is the number of cells in the stack,  $F$  is Faraday's constant, and  $i_{FC}$  is the fuel cell current [3, 4]. Therefore, increasing the current increases the hydrogen consumption rate and tends to reduce the anode pressure unless the gas flows are adjusted.

If  $T$  and  $V_{an}$  are taken as constant, the ideal-gas law and the mass balance give the pressure dynamics directly:

$$\dot{P}_{H_2} = \frac{R_{H_2}T}{V_{an}} (W_{H_2,in} - W_{H_2,out} - W_{H_2,reacted}). \quad (7)$$

This equation shows explicitly how the inlet flow, outlet flow, and hydrogen consumption affect the anode pressure.

Water transport also affects the anode side, so a simple water balance is included:

$$\dot{m}_{w,an} = W_{v,an,in} - W_{v,an,out} - W_{v,mbr}, \quad (8)$$

where  $m_{w,an}$  is the water mass in the anode,  $W_{v,an,in}$  and  $W_{v,an,out}$  are the inlet and outlet vapor mass flow rates, and  $W_{v,mbr}$  is the water transferred through the membrane [3, 4]. This balance is useful because anode humidity affects membrane hydration and, therefore, fuel cell performance.

In conclusion, this summarized anode model gives the hydrogen partial pressure and anode humidity with a small number of equations. The hydrogen pressure  $P_{H_2}$  is important because it enters the reversible-voltage expression and therefore contributes to the fuel cell voltage calculation. In addition, the pressure-dynamics equation in (7) is directly control-oriented, so  $P_{H_2}$  can later be regulated by a PI controller acting on the hydrogen inlet and outlet flows, which will be useful for the control problem developed afterward.

## 2.4. Membrane hydration model

The membrane hydration model describes how much water is stored in the membrane and how water moves through it [4]. In the reduced model, the membrane is treated as a uniform element. Its main inputs are the anode relative humidity, cathode relative humidity, stack current, and temperature, while its main outputs are the membrane water content and the membrane water flow.

Water transport through the membrane is described by two main mechanisms. The first one is electro-osmotic drag: when protons move from the anode to the cathode, they drag water with them. The corresponding molar flux is

$$N_{v,osmotic} = n_d \frac{i}{F},$$

where  $n_d$  is the electro-osmotic drag coefficient,  $i$  is the current density, and  $F$  is Faraday's constant [4]. This term increases with current and tends to move water from the anode side to the cathode side.

The second mechanism is back-diffusion. When the cathode side becomes wetter than the anode side, the water concentration gradient drives water in the opposite direction, from cathode to anode. The corresponding diffusive flux is

$$N_{v,diff} = D_w \frac{dc_v}{dy},$$

where  $D_w$  is the water diffusion coefficient and  $c_v$  is the water concentration in the membrane [4]. This term tends to oppose the drying effect created by electro-osmotic drag.

The net membrane water transport is obtained by combining these two effects:

$$N_{v,\text{membr}} = N_{v,\text{osmotic}} - N_{v,\text{diff}}.$$

The corresponding water mass flow rate is

$$W_{v,\text{membr}} = M_v A_{fc} n N_{v,\text{membr}},$$

where  $M_v$  is the molar mass of water,  $A_{fc}$  is the active area of one cell, and  $n$  is the number of cells in the stack [4]. This flow is then used in the anode and cathode water balances.

Because membrane properties depend on hydration and temperature, a dry membrane and a wet membrane do not transport water in the same way [4]. For a first simplified model, the back-diffusion term can be neglected and only the electro-osmotic drag effect is kept. In that case, the reduced membrane water-flow law becomes

$$W_{v,\text{membr}} = M_v A_{fc} n \frac{n_d i}{F}. \quad (9)$$

This approximation is often sufficient for reduced MATLAB/Simulink models, especially in early development stages [3].

In conclusion, the membrane hydration model is directly coupled to the hydrogen side because the anode humidity is one of its inputs and the membrane water flow  $W_{v,\text{membr}}$  in (9) appears in the anode water balance. It also affects the fuel cell voltage because membrane hydration changes proton conductivity, and therefore changes the ohmic voltage loss in the voltage model.

## 2.5. Oxygen model on the cathode side

Before introducing the simplifying assumption, the cathode model describes how the oxygen stored in the cathode determines the oxygen partial pressure. In the reduced model, the cathode is treated as a lumped control volume [3, 4], and the oxygen pressure is obtained from

$$P_{O_2} = \frac{m_{O_2} R_{O_2} T}{V_{ca}},$$

where  $m_{O_2}$  is the oxygen mass in the cathode,  $R_{O_2}$  is the oxygen gas constant,  $T$  is the operating temperature, and  $V_{ca}$  is the cathode volume. The oxygen concentration used in the activation-loss model is then written as [9]

$$C_{O_2} = \frac{P_{O_2}}{5.1 \times 10^6 \exp\left(-\frac{498}{T}\right)}.$$

The oxygen mass follows the balance

$$\dot{m}_{O_2} = W_{O_2,\text{in}} - W_{O_2,\text{out}} - W_{O_2,\text{reacted}},$$

with oxygen consumption given by Faraday's law,

$$W_{O_2,\text{reacted}} = \frac{M_{O_2} n}{4F} i_{FC}.$$

So, before the assumption, the cathode model can describe how current demand and air supply affect oxygen pressure and, through it, the voltage model.

The cathode water balance is also kept because water is produced at the cathode and strongly affects gas transport:

$$\dot{m}_{w,ca} = W_{v,ca,\text{in}} - W_{v,ca,\text{out}} + W_{v,mbr} + W_{v,\text{gen}},$$

with

$$W_{v,\text{gen}} = \frac{M_v n}{2F} i_{FC}.$$

### 2.5.1 Simplified cathode model with constant oxygen partial pressure

In the simplified model, the oxygen partial pressure is assumed constant as

$$P_{O_2} = P_{O_2}^* = 0.21 P_{\text{atm}}. \quad (10)$$

This means that the cathode oxygen dynamics are neglected and the oxygen mass balance is no longer needed to determine  $P_{O_2}$ . If temperature is also taken as constant, then the oxygen concentration becomes constant as well:

$$C_{O_2} = \frac{P_{O_2}^*}{5.1 \times 10^6 \exp\left(-\frac{498}{T}\right)}.$$

Therefore,  $C_{O_2}$  is no longer a state variable but a constant parameter.

Even after this assumption, the cathode humidity can still be kept dynamic through the water balance:

$$\dot{m}_{w,ca} = W_{v,ca,in} - W_{v,ca,out} + W_{v,mbr} + W_{v,gen}, \quad (11)$$

where  $W_{v,ca,in}$  and  $W_{v,ca,out}$  are the inlet and outlet vapor mass flow rates,  $W_{v,mbr}$  is the water transported through the membrane, and  $W_{v,gen}$  is the water produced by the electrochemical reaction. The generated-water term is

$$W_{v,gen} = \frac{M_v n}{2F} i_{FC}.$$

In the simplest vapor-only approximation,

$$m_{v,ca} = m_{w,ca}.$$

Then the water-vapor partial pressure and cathode relative humidity are

$$p_{v,ca} = \frac{m_{v,ca} R_v T}{V_{ca}},$$

and

$$\phi_{ca} = \frac{p_{v,ca}}{p_{sat}(T)}, \quad (12)$$

where  $R_v$  is the specific gas constant of water vapor and  $p_{sat}(T)$  is the saturation pressure of water at temperature  $T$ .

After the assumption, the effect is clear: the oxygen-related terms in the Nernst equation and in the activation-loss model become constant, so the cathode no longer contributes dynamic voltage variations through oxygen pressure. This makes the model much simpler and more suitable for control-oriented analysis, but it also means that oxygen depletion, cathode starvation, and other cathode oxygen transients are no longer captured. What remains dynamic on the cathode side is mainly the humidity through the reduced water-balance model described by (11) and (12).



## 2.6. Architecture of the reduced PEMFC model

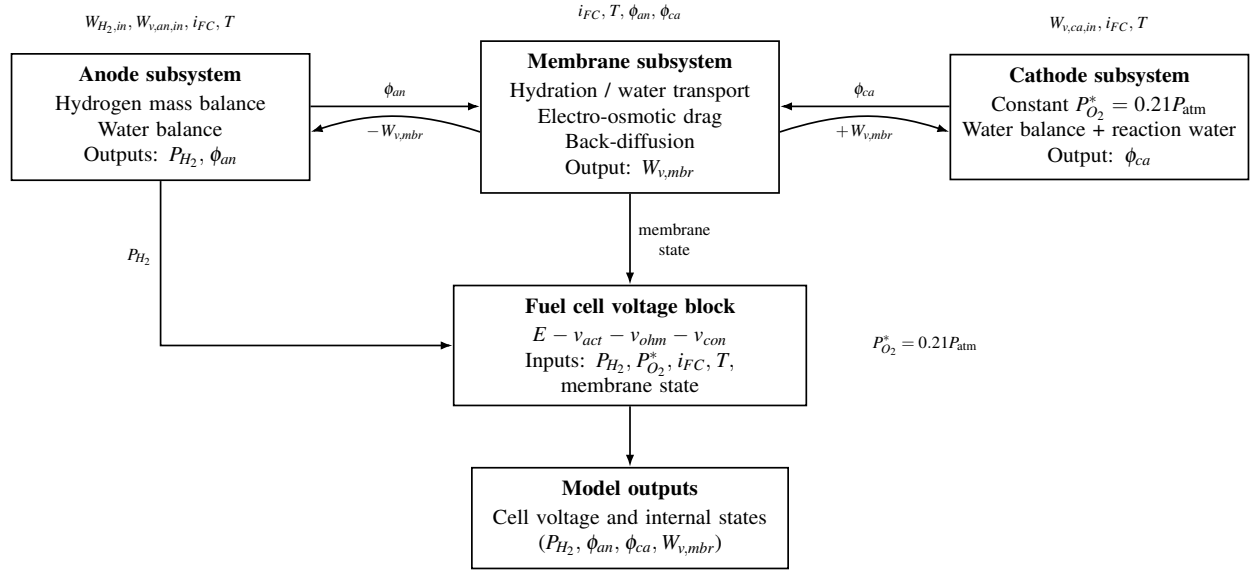


Figure 3: Block-level architecture of the reduced PEMFC model, showing the coupling between the anode, membrane, and cathode subsystems.

In summary, the reduced PEMFC model is built from three coupled subsystems: the anode, the membrane, and the cathode. The anode provides the hydrogen partial pressure and anode humidity, the membrane computes the net water transport and hydration effect, and the cathode mainly provides the cathode humidity through its reduced water balance. These variables are then combined in the voltage block to estimate the fuel cell voltage. This structure remains suitable for simulation and control design.

## 2.7. Why this model is useful

Even with the simplifying assumptions introduced above, this reduced model remains very useful for control-oriented analysis [3, 4, 9] because it still keeps the main variables that influence the stack voltage and internal water transport:

- reproducing the main dynamic trends of the stack voltage,
- studying how current demand and hydrogen-flow changes affect  $P_{H_2}$  and the voltage response,
- tracking membrane water transport and cathode humidity effects with a simple set of equations,
- designing observers and controllers, especially for hydrogen-pressure regulation and voltage-related control problems,
- tuning and validating the model parameters from measured experimental data.

## 3. Remarks on model limitations

This reduced formulation is attractive because it is compact and easy to implement in MATLAB/Simulink or for control design [3, 4]. Still, several limitations must be kept in mind [6, 7, 8]:

- channel-wise and through-plane spatial variations are ignored,
- the oxygen partial pressure is assumed constant, so cathode oxygen depletion, starvation, and oxygen transients are not captured,
- liquid-water accumulation is only represented in a very simplified manner,
- membrane transport is simplified compared with full hydration models.

So, this is a good *engineering model*, but not a complete multiphysics description of the PEMFC [6, 8].

## 4. Conclusion

This work presented the main idea of a reduced PEMFC model for control-oriented analysis [3, 4]. The modeling approach starts from the fuel-cell voltage equation and couples it with three simplified physical parts: the hydrogen dynamics on the anode side, the membrane hydration and water transport, and the cathode-side humidity dynamics. In order to keep the model simple and usable for control design, the cell is represented with lumped control volumes and the oxygen partial pressure is assumed constant. With these assumptions, the model still preserves the main interactions that determine the stack voltage, especially the effects of hydrogen pressure, membrane hydration, current demand, and cathode humidity.

Once data from the real experiments become available, the model parameters can be tuned through curve fitting between measured and simulated responses. In particular, current-voltage data and flow or pressure measurements can be used to calibrate the simplified relations so that the reduced model follows the real PEMFC behavior more closely. This calibration step will make the model more reliable for simulation, validation, and later control design.

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